Enhancing E-commerce Personalization through Big Data

A Recommendation System for Online Grocery Shopping

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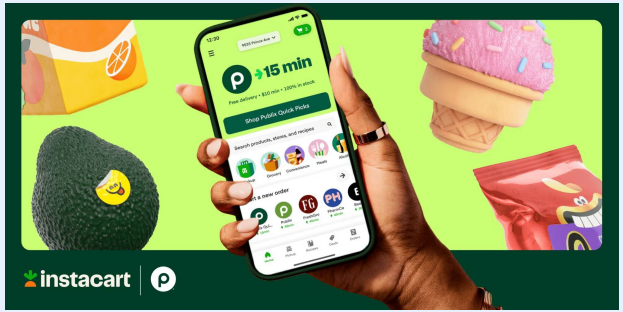
Online Grocery Industry:
Rapid Growth, Rising Expectations

- Market projected to reach $1.2 trillion by 2030 (Grand View Research, 2023)
- Customers demand convenience and personalization
- Recommendation systems now a competitive necessity

Business Problem:
The Personalization Gap in Online Grocery

Problem Statement: Online grocery shopping platforms face persistent challenges in customer retention and cart size expansion.

Opportunity: By analyzing large-scale user order data, we can identify products with high reorder rates and users with strong habitual buying behaviors.



Research Question

Can we accurately predict what a user will buy next based on past purchases?

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Prior Research

Our method tailors these advances to Instacart users' reordering behavior in grocery purchases.

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Matrix factorization (Bell & Koren, 2007)

02

Item-to-item filtering (Linden et al., 2003, Amazon)

03

Neural collaborative filtering using implicit feedback (He et al., 2017)

Goal & Objective

Build a scalable product recommendation engine using big data.

Compare three recommendation approaches:

1. Top-N Recommendation Model
2. An Alternating Least Squares (ALS)
3. Co-occurrence Model

Rationale

- Traditional tools slow at scale
- Spark enables fast in-memory computing (faster filtering, joining, and iterative machine learning training).
- Scalable recommendation algorithms, as emphasized by Bell and Koren (2007), are critical for managing massive transaction data.
- Spark’s DataFrame API and MLlib deliver scalability and high performance for datasets with millions of rows.
- Ideal for real-time recommendations

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Dataset Overview

Instacart Online Grocery Shopping 2017 (Kaggle)

Real-world, high-volume, repetitive purchase behavior includes:

- ~3.4 million orders
- 32.4+ million product-level entries
- 206,000+ unique users and 49,000+ distinct products

Six Structured CSV Files

- **orders:** Order Metadata with Time and Sequence
- **order_products_prior:** Historical Product Purchases
- **Order_products_train:** Latest Purchases order
- **products:** Product Names and IDs
- **aisles:** Aisle Categories
- **departments:** Department Categories

Tech Stack & Methodology

- Combine ALS with co-occurrence
- Cross-Sell: Bundle frequent pairs
- Reorder Automation: High-frequency items

<u>Scalable Pipeline Using PySpark</u>

- Tools: Spark SQL, MLlib, window functions
- Pipeline: Data cleaning → Feature engineering → Model training
- Evaluation Metric: Precision@10 for top product predictions

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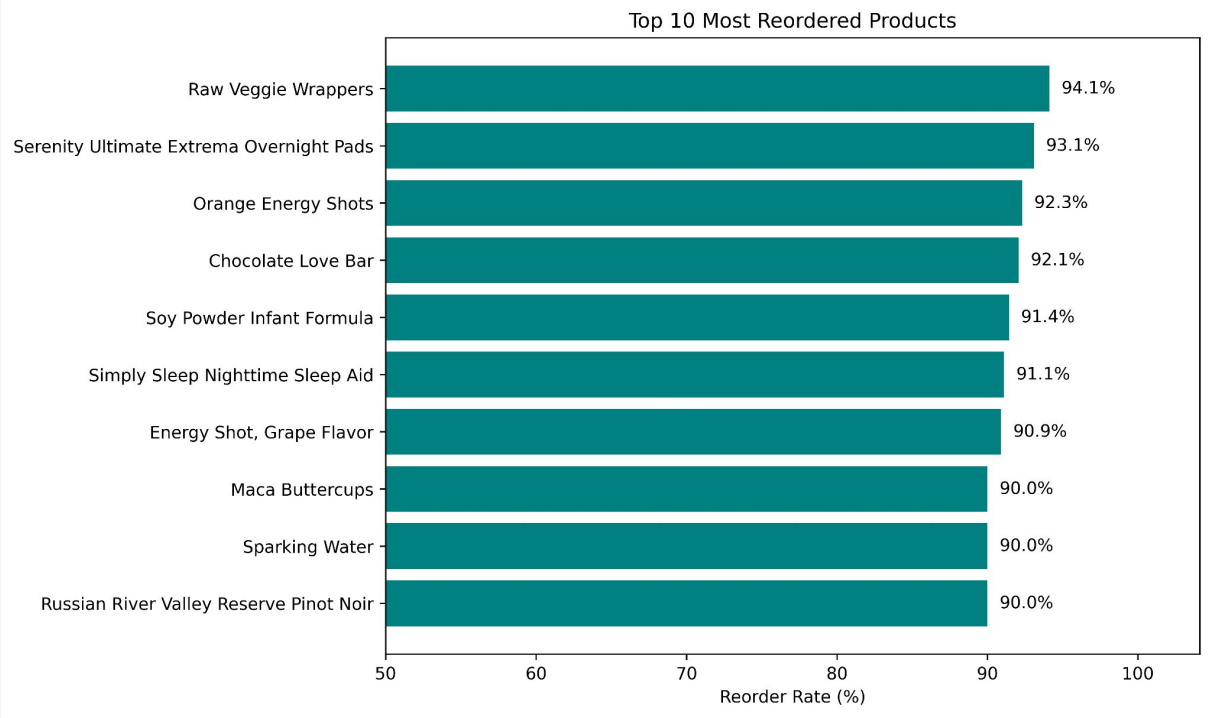
Top 10 Most Recorded Products



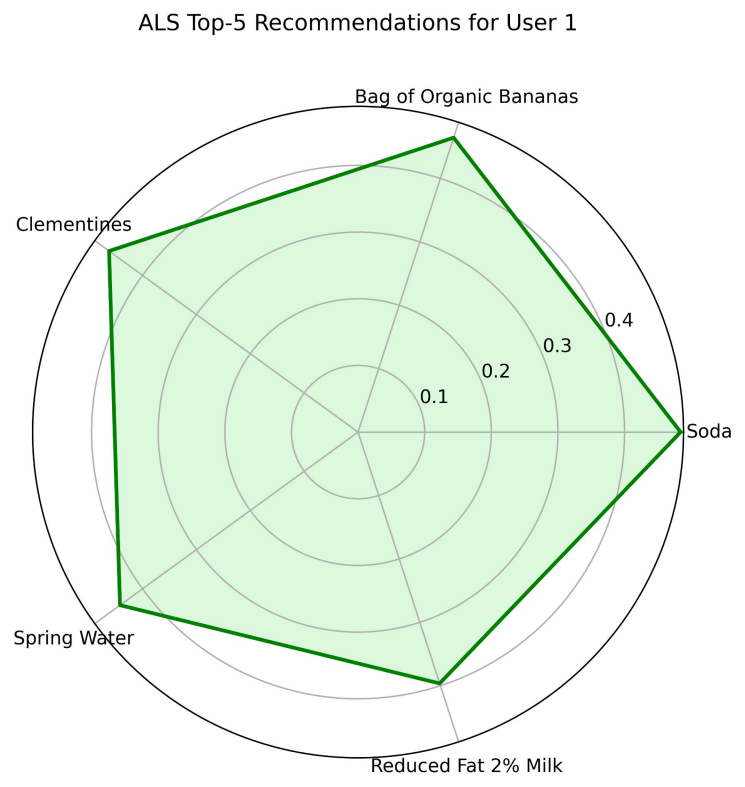
<u>Most-Ordered Product</u>

strong loyalty & habitual shopping

1. Raw Veggie Wrappers (94.1%)
2. Serenity Overnight Pads (93.1%)
3. Orange Energy Shots (92.3%)

➔ "Buy It Again" buttons
➔ Auto-refill reminders
➔ Subscription-based features

ALS Top-5 Recommendations for User

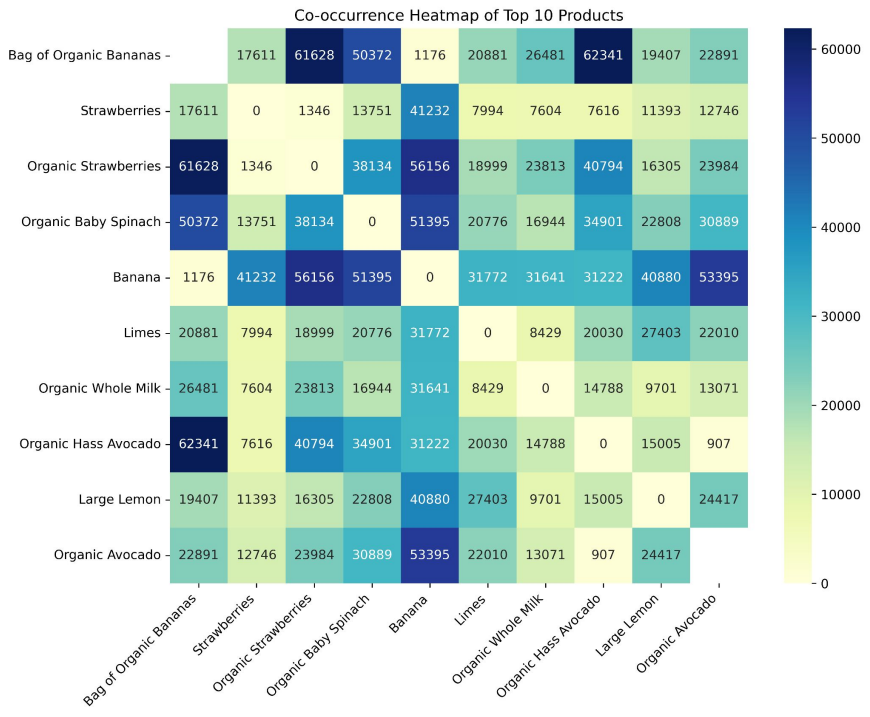


Top-5 Personalized Recommendations

1. Soda
2. Bag of Organic Bananas
3. Clementines
4. Spring Water
5. Reduced Fat 2% Milk

- ALS picks up general user preference trends.
- Weakness: may favor popular items over personalized ones.
- Limited precision, especially if a user's data is sparse.

Co-Occurrence Heatmap of Top 10 Products



<u>How often pairs of top products are bought together?</u>

Frequent combos:

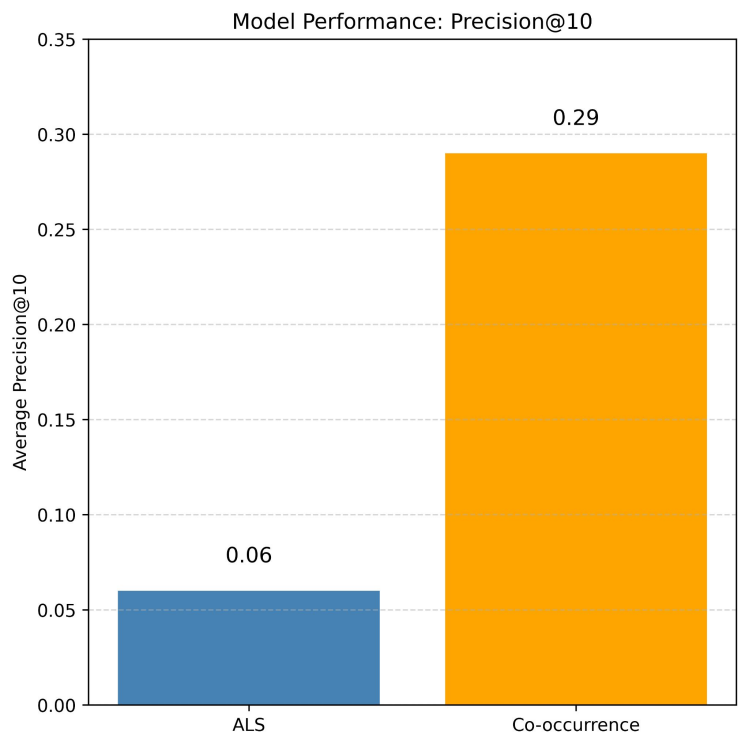
1. Bananas + Strawberries
2. Bananas + Avocados
3. Spinach + Bananas

Natural shopping patterns (e.g., smoothie ingredients).

➔ “Customers also bought" and bundle recommendations.

➔ Helps new-user recommendations based on cart contents.

Model Performance: Precision@10



ALS vs. Co-occurrence (Winner)

Precision@N = % of relevant items in the top N recommendations.

- Co-occurrence is 5x more accurate
- Co-occurrence captures frequent item pairings
- ALS suffer in sparse data (limited history).

Findings Summary

- Reorder frequency captures loyalty.
- Co-occurrence finds product pairings for upselling.
- ALS adds some personalization for users with rich histories.

➔ Together, they offer a powerful hybrid recommendation engine.

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Business Recommendation

Practical Takeaways for Online Retailers

1. Use Co-occurrence for upsell and bundle offers

2. Use Top-N Frequency for habit-based automation

3. Use ALS selectively for returning users

4. Consider building a hybrid engine to blend strengths

Conclusion

- Co-occurrence boosts upselling through smart product bundling
- ALS supports new item discovery for loyal users
- Simple models outperform in grocery settings
- Spark enables fast, scalable recommendations
- Improves user experience and sales

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Thank You！